BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: PR–IX (A-I-A8) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** PR–IX (A-I-A8) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The order of the matrix [[1], [1]] is:

(A) 2x1 (B) 1x2

(C) 1x1 (D) 2x2

Q2. The additive inverse of the matrix [[2, -1, 0], [0, -2, -3]] is:

(A) [[-2, 1, 0], [0, -2, -3]] (B) [[-2, 1, 0], [0, 2, 3]]

(C) [[2, -1, 0], [0, 2, 3]] (D) [[-2, -1, 0], [0, -2, -3]]

Q3. The adjoint of the matrix [[0, 1], [1, 0]] is:

(A) [[0, -1], [1, 0]] (B) [[0, 1], [-1, 0]]

(C) [[0, 1], [1, 0]] (D) [[0, -1], [-1, 0]]

Q4. The decimal representation of the number 50/11 is:

(A) 4.54 (B) $4.5\overline{4}$

(C) 0.22 (D) $0.2\overline{}$

Q5. The multiplicative inverse of 4.33 is:

(A) -4.33 (B) -1/4.33

(C) 1/4.33 (D) 1

Q6. The radicand in the expression $\sqrt[6]{(99x^2)}$ is:

(A) √ (B) 6

(C) 2 (D) $99x^2$

Q7. The exponential form of $\sqrt[6]{(4-y)^5}$ is:

(A) $(4-y)^{30}$ (B) $(4-y)^{6/5}$

(C) $(4-y)^{5/6}$ (D) $(4-y)^{11}$

Q8. In $(2/5)^2$, the exponent is:

(A) 2 (B) 2/5

(C) $(2/5)^2$ (D) 4/25

Q9. (4i-9)(4i) = _____.

(A) -16-36i (B) -16+36i

(C) 16+36i (D) 16-36i

Q10. The logarithmic form of $5^{-2} = 1/25$ is:

(A) $\log_5 -2 = 1/25$ (B) $\log_{-2} 5 = 1/25$

(C) $\log_{-2}(1/25) = -2$ (D) $\log_5(1/25) = -2$

Q11. The polynomial in the given expression is:

(A) $x/\sqrt{3} + 4x^2 - 8x^3$ (B) $\sqrt{x/3} + 4x^2 - 8x^3$

(C) $\sqrt{(x/3)} + 4x^2 - 8x^3$ (D) $x/\sqrt{3} + \sqrt{4x^2} - 8\sqrt{x^3}$

Q12. The zeros of the polynomial $x^2-9x-10$ are:

(A) -1 and -10 (B) 1 and 10

(C) -1 and 10 (D) 1 and -10

Q13. The L.C.M of x^2-16 and $(x+4)^2$:

(A) x+4 (B) x-4

(C) (x+4)(x-4) (D) $(x+4)^2(x-4)$

Q14. The absolute value of -12 is:

(A) 12 (B) -12

(C) -144 (D) 392

Q15. A pair of values satisfying the equation 2x+y=10 is:

(A) (4,3) (B) (3,4)

(C) (3,-4) (D) (4,-3)

SECTION-B

Time Allowed: 2 Hours 40 Minutes

Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 40 minutes.

Q2. Attempt any NINE parts. Each part carries FOUR marks.

1) Solve the following system of linear equations using Cramer's rule: $2x-y=4$, $x-2y=7$

2) If $Z_1=2+3i$ and $Z_2=3-2i$, find Z_1/Z_2.

3) Simplify with the help of Logarithm: $83\times(17.1)^2 / 29$

4) Find the value of a^2+b^2, if $(a+b)=2$ and $(a-b)=9$

5) Rationalize and simplify the expression: $5 / (5-\sqrt{5})$

6) Factorize: $49a^2c - 64b^2c$

7) Find L.C.M of $x^2-4x-12$, $x^2-9x+18$ and $x^2-10x+24$ using factorization method.

8) Solve the linear equation: $(4x-10)/4 = (x-1)/4$

9) Solve the following inequality and plot the solution on number line: $(x-4)/(x+5) > 4$; where $x \in R$

10) Using distance formula, shows that the points A(-2,-7), B(-1,-3), C(0,1) are collinear

11) Prove that: if two angles of a triangle are congruent, then the sides opposite to them are also congruent.

12) Prove that the line segment, joining the midpoints of two sides of a triangle, is parallel to the third side and is equal to one half of its length.

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 24

Note: Note: Attempt any FOUR questions. All questions carry equal marks.

Q3. Prove that any point on the bisector of an angle is equidistant from its arms.

Q4. Prove that if two sides of a triangle are unequal in length, the longer side has an angle of greater measure opposite to it.

Q5. Prove that if two triangles are similar, then the measures of their corresponding sides are proportional.

Q6. A ladder whose foot is 2.8m from the front of a house reaches a window 6.5m above the ground. Calculate the length of the ladder.

Q7. Prove that parallelogram on the same base and lying between the same parallel lines (or of the same altitude) are equal in area.

Q8. Construct a triangle ABC, given that mAB = 5cm, mBC = 5cm and m∠A = 60°.